

Bhukya Roopak Krishna

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Professional Summary

ML Engineer and Full Stack AI Developer with real-world experience **building, fine-tuning, and deploying end-to-end machine learning models** in production environments. Skilled at integrating AI systems into full-stack web applications. Hands-on expertise in **LLM fine-tuning, MLOps pipelines, Agentic AI, and Generative AI**, with a proven ability to deliver scalable AI-powered products from model development through to deployment.

Education

B.Tech, Materials Science & Metallurgical Engineering
Indian Institute of Technology (IIT) Hyderabad

2021 – 2025
CGPA: 7.34 / 10

Technical Skills

Languages:	Python, C++, JavaScript, TypeScript
ML / DL:	PyTorch, Scikit-Learn, HuggingFace, Transformers, PEFT/LoRA, LLM Fine-Tuning, LangChain, RAG, NLP
Data & Science:	NumPy, Pandas, FAISS, Jupyter Notebook
MLOps & Cloud:	Kubeflow, Kubeflow Pipelines, MLflow, Elyra, Docker, AWS
Web Frameworks:	React, Next.js, FastAPI, Flask, REST APIs
Databases & Tools:	PostgreSQL, MySQL, Git, GitHub, Postman

Professional Experience

Full Stack AI Developer

Net Connect Global Pvt. Ltd., Onsite – Hyderabad

Nov 2025 – May 2026

- Deployed a production-grade **AI interview proctoring system** using computer vision (face detection, gaze tracking, mobile detection), audio anomaly detection, and keyboard monitoring — generating **30+** automated proctoring reports daily at **93% detection accuracy** across all modalities.
- Reduced manual identity validation effort significantly by engineering an automated **KYC pipeline** using Aadhaar/PAN OCR extraction and facial similarity models, processing **1,000+** candidate verifications with **97% face-match accuracy**.

Machine Learning Intern

Mobius Networks Pvt. Ltd., Onsite – Hyderabad

Jun 2025 – Aug 2025

- Improved **knowledge graph link prediction by 33%** by replacing R-GCN with **CompGCN** models and deploying a reusable end-to-end **MLOps workflow** on Kubeflow Pipelines covering GNN training, evaluation, and versioning.
- Fine-tuned **Mistral** and other LLMs on proprietary domain-specific datasets using **PEFT** techniques, achieving a **1.4% gain** in task performance while reducing inference latency by **0.3 seconds**.

Projects

Options Trading System – Arbitrage Detection

Jan 2026 – Apr 2026

- Achieved **71% directional accuracy** on historical market data by building an **ML-powered quantitative trading system** that models volatility surfaces across strikes and expiries for options market analysis.
- Reached **0.84 F1-score** in arbitrage opportunity detection by designing **pricing-discrepancy detection algorithms** over options chains, integrated into an end-to-end data processing and visualization pipeline.

Certificate Verification Platform (SkillKendra)

May 2025 – Jul 2025

- Enabled **fraud detection across 50+ certification providers** by building a **multi-agent pipeline** combining OCR-based text extraction and automated validation workflows, achieving **87% OCR extraction accuracy**.
- Increased system reliability by implementing **certificate ID cross-verification** against official records with candidate name-matching, plus a manual fallback for low-confidence OCR cases — reducing false rejection rates.

Innovative Ranking Techniques in Undirected Networks

Mar 2025 – May 2025

- Developed novel graph ranking algorithms addressing dangling-node challenges using **Markov chain** and **PageRank** methodologies on undirected network structures, handling graphs with **10K+ dangling nodes**.
- Introduced two original pre-filtering methods – Context Sharing and Complementary Networks – improving ranking accuracy and robustness for sparse graph topologies.

Achievements & Activities

- Solved 300+ DSA problems on LeetCode, GeeksforGeeks, and Codeforces; active NSS member contributing to Clean India Drives, Cancer Awareness Walkathons, and plantation drives.
- Traveled across 25 states and 4 Union Territories in India and 3 countries internationally.